

## TASKS BASED ON NEW YORK CITY AIRBNB DATASET

### Program:

# (Task 1: Using Data Cleaning (Missing Values, Price Outliers, Duplicate Listings))

```
import pandas as pd
```

```
# Load Dataset
```

```
df = pd.read_csv(r"C:\Users\Alana\Downloads\AB_NYC_2019.csv.zip")
```

```
# Remove extra spaces from column names
```

```
df.columns = df.columns.str.strip()
```

```
# Check missing values
```

```
print("Missing Values")
```

```
print(df.isnull().sum())
```

```
# Fill missing values
```

```
df["name"] = df["name"].fillna("Unknown")
```

```
df["host_name"] = df["host_name"].fillna("Unknown")
```

```
df["reviews_per_month"] =
```

```
df["reviews_per_month"].fillna(df["reviews_per_month"].median())
```

```
# Remove duplicate listings
```

```
print("Duplicate Rows Before:", df.duplicated().sum())
```

```
df = df.drop_duplicates()
```

```
print("Duplicate Rows After:", df.duplicated().sum())
```

```
# Remove price outliers using IQR
```

```

Q1 = df["price"].quantile(0.25)
Q3 = df["price"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df = df[(df["price"] >= lower) &
        (df["price"] <= upper)]

print(df.info())

# Save cleaned dataset
df.to_csv("airbnb_clean.csv", index=False)

```

## Output:

Missing Values

|                     |    |
|---------------------|----|
| id                  | 0  |
| name                | 16 |
| host_id             | 0  |
| host_name           | 21 |
| neighbourhood_group | 0  |
| neighbourhood       | 0  |
| latitude            | 0  |
| longitude           | 0  |
| room_type           | 0  |
| price               | 0  |
| minimum_nights      | 0  |

```
number_of_reviews      0
last_review             10052
reviews_per_month       10052
calculated_host_listings_count  0
availability_365        0
```

dtype: int64

Duplicate Rows Before: 0

Duplicate Rows After: 0

<class 'pandas.core.frame.DataFrame'>

Index: 45923 entries, 0 to 48894

Data columns (total 16 columns):

| #  | Column                         | Non-Null Count | Dtype   |
|----|--------------------------------|----------------|---------|
| 0  | id                             | 45923 non-null | int64   |
| 1  | name                           | 45923 non-null | object  |
| 2  | host_id                        | 45923 non-null | int64   |
| 3  | host_name                      | 45923 non-null | object  |
| 4  | neighbourhood_group            | 45923 non-null | object  |
| 5  | neighbourhood                  | 45923 non-null | object  |
| 6  | latitude                       | 45923 non-null | float64 |
| 7  | longitude                      | 45923 non-null | float64 |
| 8  | room_type                      | 45923 non-null | object  |
| 9  | price                          | 45923 non-null | int64   |
| 10 | minimum_nights                 | 45923 non-null | int64   |
| 11 | number_of_reviews              | 45923 non-null | int64   |
| 12 | last_review                    | 36911 non-null | object  |
| 13 | reviews_per_month              | 45923 non-null | float64 |
| 14 | calculated_host_listings_count | 45923 non-null | int64   |
| 15 | availability_365               | 45923 non-null | int64   |

dtypes: float64(3), int64(7), object(6)

memory usage: 6.0+ MB

None

### **Program:**

```
# (Task 2:Using Data Analysis )
```

```
import pandas as pd
```

```
# Load cleaned dataset
```

```
df = pd.read_csv(r"C:\Users\Alana\Downloads\AB_NYC_2019.csv .zip")
```

```
# Most Expensive Neighbourhoods
```

```
print("Most Expensive Neighbourhoods")
```

```
print(df.groupby("neighbourhood_group")["price"].mean().sort_values(ascending=False))
```

```
# Room Type Comparison
```

```
print("\nRoom Type Comparison")
```

```
print(df.groupby("room_type")["price"].mean())
```

```
# Availability Analysis
```

```
print("\nAvailability Analysis")
```

```
print(df["availability_365"].describe())
```

```
# Reviews vs Pricing
```

```
print("\nReviews vs Price Correlation")
```

```
print(df[["number_of_reviews","price"]].corr())
```

### **Output:**

Most Expensive Neighbourhoods

#### neighbourhood\_group

Manhattan 196.875814

Brooklyn 124.383207

Staten Island 114.812332

Queens 99.517649

Bronx 87.496792

Name: price, dtype: float64

#### Room Type Comparison

##### room\_type

Entire home/apt 211.794246

Private room 89.780973

Shared room 70.127586

Name: price, dtype: float64

#### Availability Analysis

count 48895.000000

mean 112.781327

std 131.622289

min 0.000000

25% 0.000000

50% 45.000000

75% 227.000000

max 365.000000

Name: availability\_365, dtype: float64

#### Reviews vs Price Correlation

number\_of\_reviews price

number\_of\_reviews 1.000000 -0.047954

price                    -0.047954   1.000000

### **Program:**

# (Task 3: Using Data Visualization (Price Distribution, Neighbourhood Comparison, Correlation Heatmap, Scatter Plot)

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Load cleaned dataset
```

```
df = pd.read_csv(r"C:\Users\Alana\Downloads\AB_NYC_2019.csv.zip")
```

```
# Price Distribution
```

```
plt.figure(figsize=(8,5))
```

```
sns.histplot(df["price"], bins=40, kde=True)
```

```
plt.title("Price Distribution")
```

```
plt.show()
```

```
# Neighbourhood Comparison
```

```
plt.figure(figsize=(8,5))
```

```
sns.barplot(data=df,
```

```
          x="neighbourhood_group",
```

```
          y="price")
```

```
plt.xticks(rotation=45)
```

```
plt.title("Neighbourhood Comparison")
```

```
plt.show()
```

```
# Scatter Plot
```

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(data=df,
```

```

        x="number_of_reviews",
        y="price",
        hue="room_type")
plt.title("Reviews vs Price")
plt.show()

# Correlation Heatmap
plt.figure(figsize=(10,7))
sns.heatmap(df.corr(numeric_only=True),
            annot=True,
            cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

```

## Output:



